Scalability & Future Plan

Date	25 September 2026
Team ID	SWTID-2026-4307
Project Name	AI Weather Wise API
Maximum Marks	1 Mark

Scalability & Future Plan:

This document describes how the AI Weather Wise API can be scaled to support more users, larger weather-data requests, improved performance, stronger security, and additional AI-powered features in the future.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Depends on external weather and AI APIs.	Service availability and response time depend on external providers.	High
2	Limited capacity for very large numbers of simultaneous users.	High traffic may increase response time.	Medium
3	Basic data storage and caching in the current version.	Repeated requests may increase API usage and database load.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Suitable for small/medium usage.	Use load balancing, horizontal scaling, and rate limiting.
2	Data Storage	MongoDB stores user and project data.	Use indexed collections, optimized queries, backup, and scalable cloud storage.
3	Performance	Real-time API calls are made for weather information.	Add caching, asynchronous processing, and API response optimization.
4	Security	JWT authentication protects API access.	Add token expiry/refresh, validation, HTTPS, stronger access controls, and monitoring.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Weather alerts and location-based notifications	1–2 Months	Faster and more useful weather updates for users.
Phase 2	Advanced AI recommendations and personalized insights	3–4 Months	More relevant weather-based guidance.
Phase 3	Caching, cloud deployment, and scalable infrastructure	5–6 Months	Improved performance and support for higher user load.
Phase 4	Mobile/web dashboard, analytics, and multilingual support	6–12 Months	Wider accessibility and better user experience.

Future Enhancement Summary

The AI Weather Wise API can be extended with real-time alerts, personalized AI insights, cloud scalability, caching, analytics, mobile/web integration, and multilingual support while maintaining secure API access.